Management of patient with Deep venous thrombosis

Learning objectives

- Plan effective care of patient with deep vein thrombosis
- Utilize critical thinking skills and clinical competences needed when providing nursing care to patient with deep vein thrombosis
- Critically analyzed the causes, pathologic changes, clinical manifestation for patient deep vein thrombosis
- Synthesize clinical evidence to solve problems related to the management of patient care with deep vein thrombosis
- Design and implement the nursing care plan and health teaching for patients with deep vein thrombosis

- Conduct appropriate nursing activities skillfully and in accordance with best evidence-based practice for the patients with deep vein thrombosis
- Manage time effectively and set priorities.
- Measure critically the outcomes of nursing activities.
- Use problem- solving skills.
- Apply communication skills effectively with medical staff in inter-professional, social and therapeutic context
- Works effectively with a team

Teaching and Learning Method

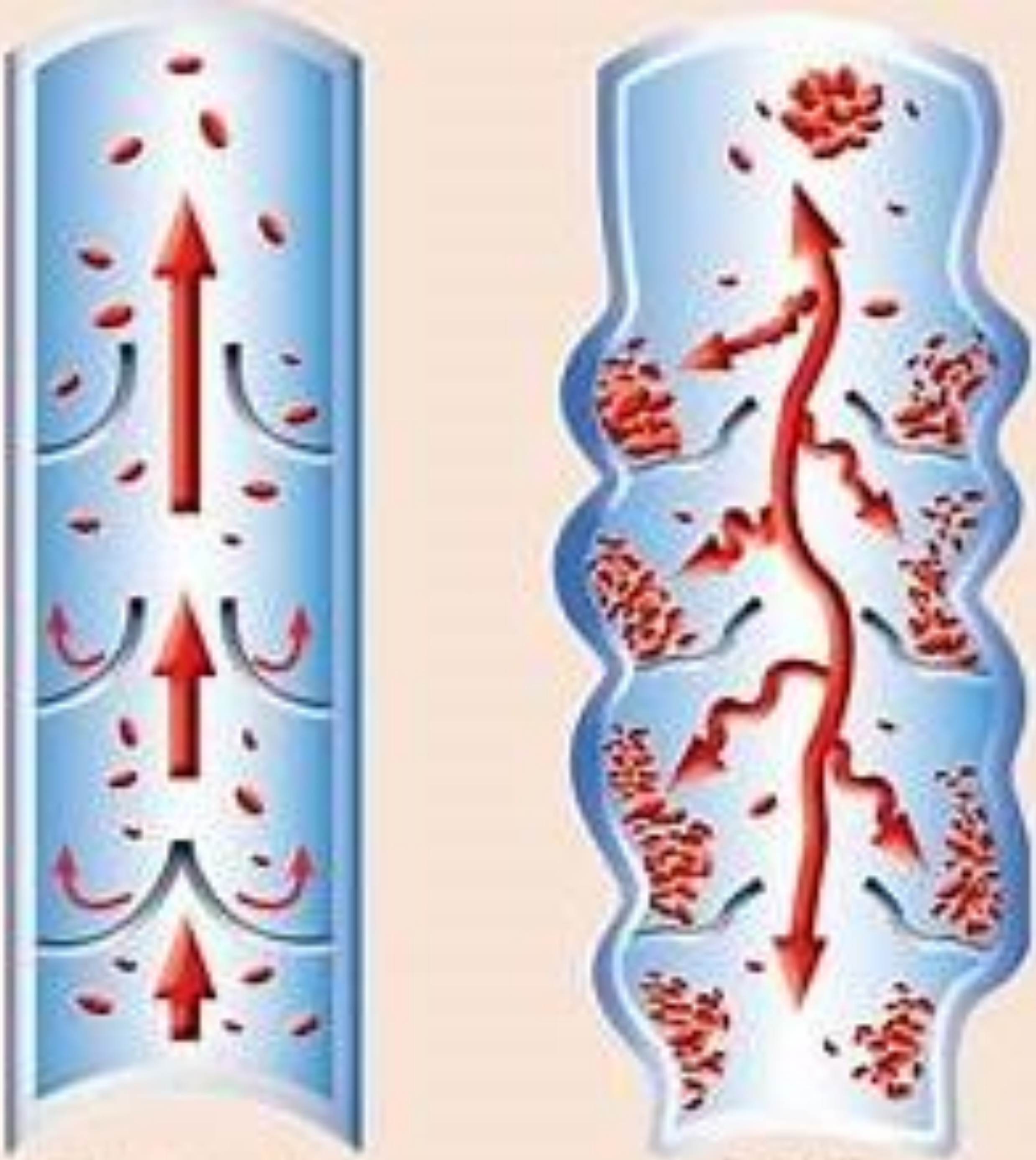
- Lecture
- Discussion
- Written assignments
- Case study

Outlines:

1. Introduction
2. Pathophysiology
3. Risk factors
4. Clinical manifestation
5. Complications
6. Diagnosis
7. Medical management
8. Surgical management
9. Nursing management

Introduction

Deep vein thrombosis (DVT) is an obstructive disease hindering venous reflux mechanism. DVT usually involves the lower limb venous system, with clot formation originating in a deep calf vein and propagating proximally.



Normal

DVT


HEALTHADVIC
DVT
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Deep Vein Thrombosi
Causes, Symptoms and Treatment

Thrombophlebitis: is an inflammation of the vein walls.

- **Phlebothrombosis:** a thrombus develops initially in the veins as a result of stasis or hypercoagulability but without inflammation.

Definition

Deep vein thrombosis (DVT) is defined as development of thrombosis within the deep veins of the pelvis or lower limbs.

Risk factors

Although the exact cause of DVT remains unclear, three factors, known as Virchow’s triad, are believed to play a significant role in its development.

Endothelial Damage

- Trauma
- Surgery
- Pacing wires
- Central venous catheters
- Dialysis access catheters
- Local vein damage
- Repetitive motion injury

Venous Stasis

- Bed rest or immobilization
- Obesity
- History of varicosities
- Spinal cord injury
- Age (greater than 65 years)

Altered Coagulation

- Cancer
- Pregnancy
- Oral contraceptive use
- Hyperhomocysteinemia
- Elevated factors II, VIII, IX, XI
- Antithrombin III deficiency
- Polycythemia
- Septicemia

Clinical manifestations

The clinical manifestations of DVT vary considerably between patients and can be:

- symptomatic or asymptomatic,
- bilateral or unilateral,
- severe or mild.
- Edema of the effected extremities is a common specific symptom of DVT. Thrombus that do not completely obstruct the venous outflow are often asymptomatic.

- **Pain:** There is usually aching discomfort and tightness in the involved calf or thigh, which are aggravated by muscular exercise.
- Dilation of the superficial veins,
- Redness or discoloration of the skin (Bluish),
- Warmth of the affected area

- ❖Tenderness in the calf this is important signs
- ❖The calf may feel tight, heavy or pulled
- ❖functional impairment, ankle engorgement

Complications of deep vein thrombosis

1. **Pulmonary embolism (PE)** is the most immediate and significant complication of DVT.

- ✓ This can cause **cardiogenic shock** followed by circulatory failure and death.
- ✓ This can cause chest pain, shortness of breath or coughing.
- ✓ PE can happen hours or even days after the DVT has formed and may occur when there have been no obvious signs of a DVT.

So the nurse should assess continuously for neurological defects or cardiopulmonary function as (fainting, tachycardia, seizure, headache, speech- thinking- walking problems, week limb, chest pain, abnormal breathing)

Recurrent DVT

occurs within a year after the initial event; because of persistent abnormalities in effected vasculature after the initial DVT.

Post-thrombotic syndrome (PTS) The primary symptoms of PTS include pain, varicose veins, edema, venous ectasia **(distention or dilation of a vessel**), induration, and ulceration.

- **Chronic venous insufficiency (regurgitation)** long term DVT can degenerate the valves in veins that control blood flow. It can result from valvular destruction, allowing retrograde flow of venous blood.

Diagnosis of Deep vein thrombosis

1. ***Coagulation profile***

- A prolonged prothrombin time **(PT)** or activated partial thromboplastin time **(APTT)** may not indicate a low probability of DVT.

2. ***D-dimer***

3. ***Venous ultrasonography***

4. ***Duplex Doppler ultrasound***

5. ***Magnetic resonance imaging (MRI).***

6. ***Contrast venography.***

7. ***Computed tomography venography (CTV)***

Prevention

- Application of elastic compression stockings.
- The use of intermittent pneumatic compression devices.
- Special body positioning and exercise.
- Unfractionated or low molecular weight heparin.

Goals of DVT management

There is more than one goal of treatment for deep vein thrombosis (DVT). The goals include:

- Preventing a clot from growing
- Preventing a clot from breaking off and traveling to the lung or other organ
- Avoiding long-lasting complications, such as leg pain and swelling
- Preventing blood clots from recurring

Medical management

❖ Blood Thinners for DVT

Blood thinners (**also called anticoagulants**) The two main types of anticoagulants are heparin and warfarin (Coumadin).

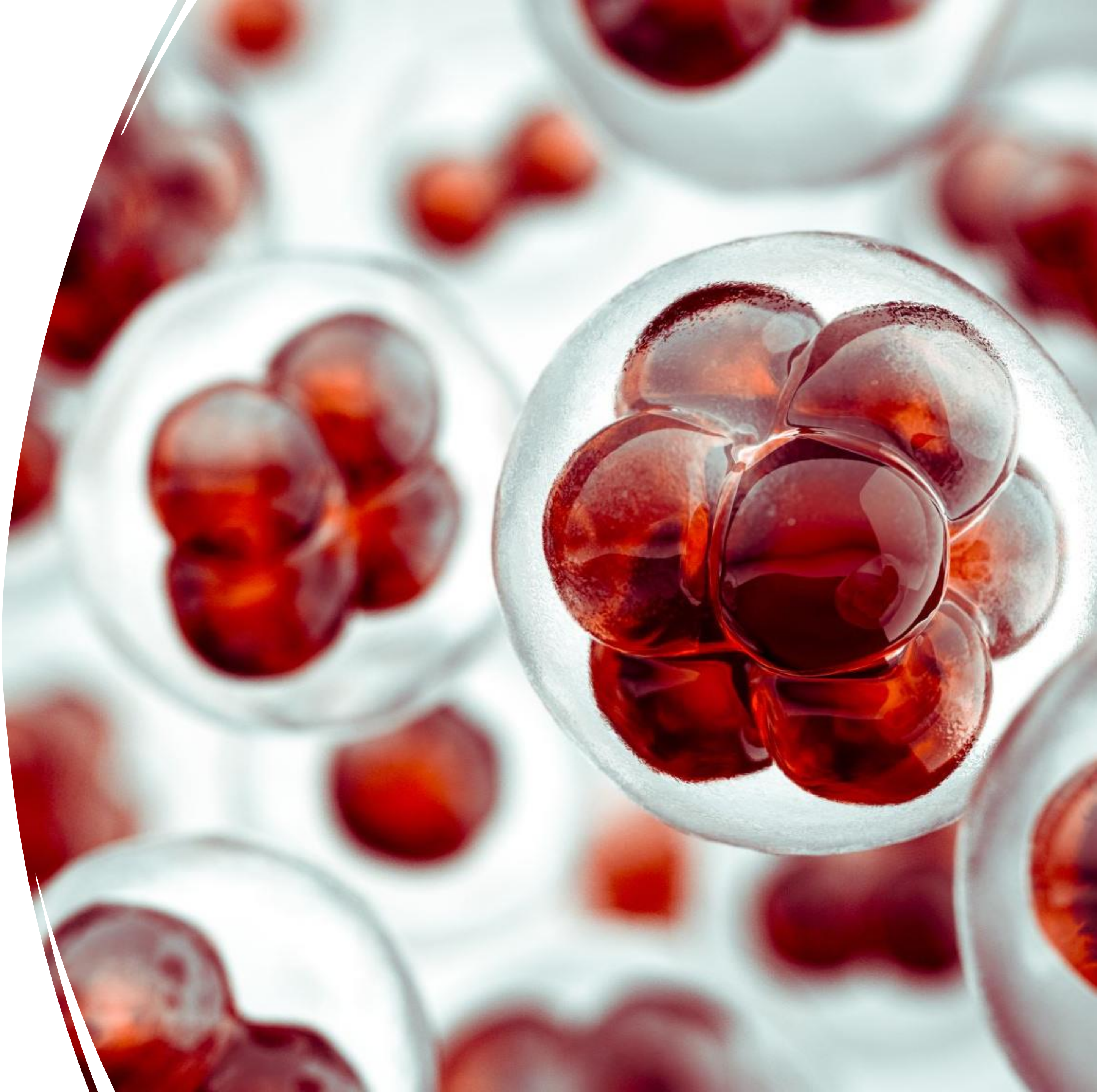
Blood thinners can:

- Keep a clot from growing or breaking off
- Prevent new clots from forming

1. Heparin.

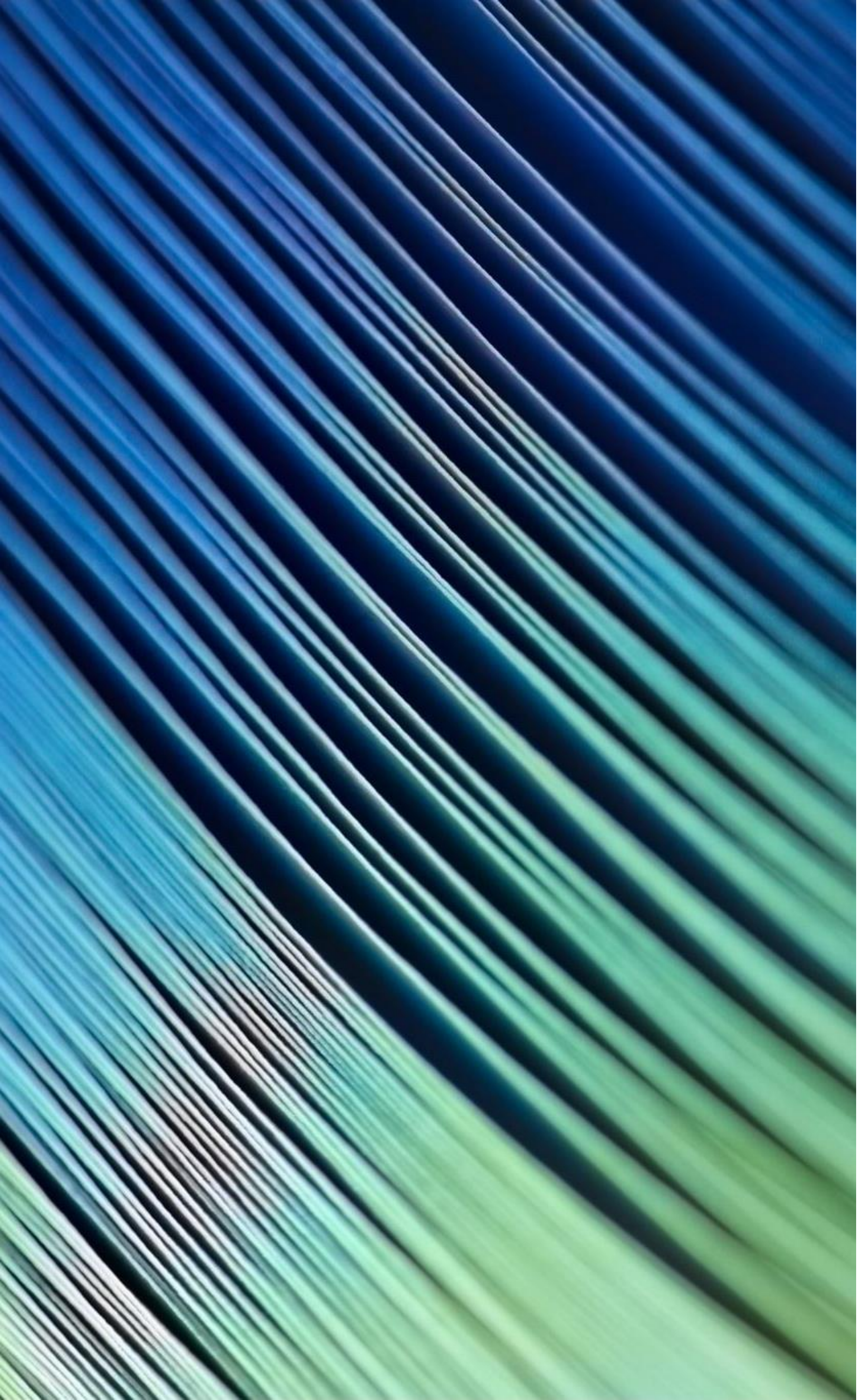
2-Oral anticoagulants (Warfarin).

3-Thrombolytic Therapy.



Surgical management

- **Venous thrombectomy**. In very rare cases, surgery is required to remove a deep vein clot. This may be true if having a severe type of DVT that does not respond well to no surgical DVT treatment.



Catheter-directed thrombolysis

- Catheter-directed thrombolysis rapidly breaks up a clot, restoring blood flow. It may also preserve valve function in the vein that contained the clot. The procedure is done in the hospital and carries a higher risk of bleeding problems and stroke than does anticoagulant therapy.

- **Inferior Vena cava filter**

This small metal device is temporarily inserted to capture blood clots and prevent them from moving to other areas of the body. The filter allows blood to pass through the vein as it normally would.

Nursing Management

Elevation and compression: Elevating the affected leg may help reduce symptoms of DVT, such as swelling and pain.

compression stockings to reduce the risk of recurrence.

Anticoagulant therapy, the nurse must frequently monitor the coagulation tests PT, PTT, hemoglobin and hematocrit, platelet count, and fibrinogen level.

Close observation to **detect bleeding;** if bleeding occurs, it must be reported immediately and anticoagulant therapy discontinued.

- **Bed rest**, elevation of the affected extremity,
- Elastic compression **stockings, and analgesics for pain** relief are adjuncts to therapy. They help to improve circulation and increase comfort.
- **Warm, moist packs** applied to the affected extremity **reduce the discomfort** associated with deep vein thrombosis, as do **mild analgesics** prescribed for **pain control**. When the patient begins to ambulate, elastic **compression stockings** are used.
- **Walking is better than standing or sitting** for long periods. **Bed exercises.**

Monitoring and managing potential complications

1. **Bleeding:**
2. **Thrombocytopenia:** decrease in platelets which may develop in patients who receive heparin for more than 5 days or on re-administration after a brief interruption of heparin therapy.

- ✓ **Drug Interactions:** oral anticoagulants interact with many other medications and herbal and nutritional supplements.
- ✓ Close monitoring of the patient’s medication schedule is necessary.
- ✓ Medications and supplements that potentiate oral anticoagulants include salicylates, anabolic steroids, green tea, and vitamin E; those that decrease the anticoagulant effect include phenytoin, barbiturates, diuretics, estrogen, and vitamin C.

- **Providing comfort:** bed rest
- Elevation of the affected extremity,
- Elastic compression stockings,
- And analgesics for pain relief are adjuncts to therapy to improve circulation and increase comfort.
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- When the patient begins to ambulate, elastic compression stockings are used.

Any Questions